Status	Finished
Started	Friday, 16 October 2026, 8:39 PM
Completed	Friday, 16 October 2026, 8:43 PM
Duration	3 mins 59 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It avoids the need for recursion
b. It simplifies the sorting process by using only one pass
c. It only works on small datasets
d. It allows for efficient parallel processing and sorting of data

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **2**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Bubbling up the largest element to its correct position with each pass
b. Sorting data in a non-sequential manner
c. Being the most efficient sorting algorithm
d. Using the divide-and-conquer approach

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Binary Search
b. Bubble Sort
c. Merge Sort
d. Linear Search

The correct answer is: Merge Sort

Question **4**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Duplicates
b. Searching
c. All of the above
d. Selection

The correct answer is: All of the above

Question **5**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is difficult to implement
b. Bubble Sort is less efficient for large lists
c. Bubble Sort cannot handle duplicate elements
d. Bubble Sort does not guarantee sorted order

The correct answer is: Bubble Sort is less efficient for large lists

Question **6**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It is based on the divide-and-conquer approach
b. It does not require recursion
c. It sorts data using a single pass
d. It is less efficient than bubble sort

The correct answer is: It is based on the divide-and-conquer approach

Question **7**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. sorted()
b. arrange()
c. order()
d. sort()

The correct answer is: sort()

Question **8**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It makes the search operation slower
b. It increases the number of comparisons needed
c. It has no effect on the search operation
d. It allows for faster searching

The correct answer is: It allows for faster searching

Question **9**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Greedy method
b. Dynamic programming
c. Backtracking
d. Divide and conquer

The correct answer is:
Divide and conquer

Question **10**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It allows for quicker identification of duplicates
- b. It makes the analysis slower
- c. It complicates the analysis process
- d. It has no effect on the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **11**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

- a. Quick sort
- b. Insertion sort
- c. Heap sort
- d. Bubble sort

The correct answer is:
Insertion sort

Question **12**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Sorting data in a single pass
- b. Sorting data sequentially
- c. Rearranging data without sorting
- d. Dividing the input into parts, solving each part, and combining the solutions

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **13**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Makes data harder to manage
b. Makes code execution slower
c. Provides a basis for other algorithms to work efficiently
d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting decreases the efficiency of selection operations
b. Sorting is rarely used in programming
c. Sorting makes it harder to search for items
d. Sorting helps in finding duplicates quickly

The correct answer is: Sorting helps in finding duplicates quickly

Question **15**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It divides the list into sublists
b. It bubbles up the largest element in each pass
c. It recursively sorts subproblems
d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass